

Derrin Bright

Bioinformatics | Biotechnology | ML in Biology

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Education

National University of Singapore (NUS) Master of Science in Precision Health and Medicine, Yong Loo Lin School of Medicine	Singapore Aug 2026-Present
Vellore Institute of Technology (VIT) Bachelor of Technology in Biotechnology	CGPA: 8.8 Vellore, India 2022-2026

Technical Skills

Computational Biology Tools	BLAST AlphaFold AutoDock Vina GROMACS MGLTools Bioconductor Open-Babel Cytoscape Seurat GSEA FIMO DESeq2 WGCNA MSigDB limma
Data Analysis Tools	Neo4J KNIME NetworkAnalyst TCGA/GDC
Programming Tools	R Python
Machine Learning Tools	TensorFlow Keras NumPy Pandas Matplotlib scikit-learn XGBoost
Certificates	Machine Learning A-Z Python for Data Science, AI & Development Bulk RNA-Seq Data Analysis Advanced Bioinformatics Single-Cell RNA-Seq Data Analysis using R & Python

Research Experiences

- Project Intern at Nanyang Technological University, Singapore** Jan 2026 - June 2026
- Replicated a transcription-factor activity signature for Alzheimer's disease across three independent post-mortem brain cohorts (637 donors), with 21 of 23 discovery factors matching direction of effect (permutation $p = 1 \times 10^{-4}$).
 - Found that a public metadata file had not been updated after a 2025 correction, leaving 9% of donors with incorrect pathology staging, and rebuilt the cohort from the corrected source before analysis.
 - Supervised by Dr. Guillaume Thibault | Guillaume Thibault Lab
- Research Intern at Boston University, USA** Sept 2025 - April 2026
- Scanned Long Control Region sequences across 207 HPV types and identified 104 transcription factors with significantly different binding site densities between high-risk and low-risk types.
 - Integrated results with TCGA transcriptomics and public ChIP-seq data, finding over half of the high-risk-enriched TFs expressed in HPV-positive tumors and confirming LCR occupancy for seven of them; first author on the resulting paper in *Viruses* (2026).
 - Supervised by Dr. Juan Fuxman Bass
- Research Intern at University of Liverpool, UK** Sept 2025 - August 2026
- Built and released 30 genome-scale metabolic models across five immune cell types spanning CLL, Richter Transformation and healthy donors, reconstructed from 1,500 single-cell bootstrap models using ftINIT and GECKO 3 enzyme constraints.
 - Identified a one-way metabolic expansion in Richter Transformation (21 capabilities gained, none lost), with de novo fatty-acid synthesis dominating the gains.
 - Audited flux reproducibility across 1,385 exchange reactions, found none uniquely determined by the models, and re-scoped the analysis to capability and network structure rather than flux magnitude.
 - Supervised by Dr. Francesco Del Carratore
- Research Intern at IIT Bombay, India** May 2025 - Sept 2026
- Performed differential expression analysis and WGCNA across four autoimmune diseases (SLE, rheumatoid arthritis, type 1 diabetes, ankylosing spondylitis) to identify disease-associated co-expression modules and hub genes, followed by GSEA and PPI network construction. Contributed to a manuscript under review at *Journal of Autoimmunity*.
 - Built an independent DESeq2 and WGCNA pipeline on 1,030 TCGA breast cancer tumours to compare PAM50 molecular subtypes, identifying 8,075 differentially expressed genes and 35 co-expression modules, with hub genes extracted from the four modules most strongly associated with subtype.
 - Ran Gene Set Enrichment Analysis against the MSigDB Hallmark collection, recovering all 50 gene sets and confirming proliferation signalling in basal and HER2 tumours against oestrogen response signalling in luminal tumours.
 - Supervised by Dr. Prakriti Tayalia
- Research Intern at Ashoka University, India** Jan 2025 - Mar 2025
- Analyzed five assembled *E. coli* genomes for carbapenem and beta-lactam resistance using BLAST and RGI-CARD to identify mutations, and correlated genotype-phenotype relationships through AST report analysis.
 - Performed Multiple Sequence Alignment (MSA) of Penicillin-Binding Proteins (PBPs) from the five *E. coli* genomes against a reference wild-type sequence to detect conserved mutations, and conducted genomic analysis that led to the identification of a 12-base pair insertion in the PBP3 gene, contributing to antibiotic resistance mechanisms.
 - Automated genomic data processing by developing a custom script to systematically rename and organize FASTA files, creating a structured dataset for phylogenetic analysis using the PhaME tool.
 - Supervised by Dr. Shraddha Karve
- Biotech Team Lead at NovaLinks, India** Feb 2025 - April 2025
- Led a project integrating miRNA analysis and histopathological imaging with machine learning to enhance lung cancer subtype classification and improve diagnostic precision.
 - Conducted research on miRNA biomarkers specific to lung cancer and provided biological insights to support the development of an accurate classification model for distinguishing adenocarcinoma from squamous cell carcinoma.

Publications

Comparative Analysis of Transcription Factor Binding Sites in the Long Control Region Across Human Papillomavirus Types | *Viruses*, 2026, 18(6):646 (MDPI)

Integrative transcriptomic analysis identifies predictive gene signatures across autoimmune diseases: implications for personalized therapeutic targeting | *Under review at Journal of Autoimmunity (Elsevier)*

Comparative Analysis of HPV-Positive Cervical and Head and Neck Cancers Using WGCNA and Machine Learning Identifies Shared Biomarkers | *Under review at Tumour Virus Research (Elsevier)*

HPV-Associated Transcriptional Landscape in Cervical and Head and Neck Cancers | *In revision at Infectious Agents and Cancer (Springer Nature)*

Technical Projects

Molecular Subtype Comparison in Breast Cancer Using Co-Expression Networks and GSEA

- Compared PAM50 basal and HER2 tumours against luminal A and B tumours using 1,030 TCGA-BRCA samples and 37,048 genes, with DESeq2 identifying 8,075 differentially expressed genes.
- Built a signed WGCNA network on 18,667 genes, detecting 35 modules and selecting hub genes by module membership and gene significance thresholds.
- Ran GSEA against MSigDB Hallmark gene sets with 1,000 phenotype permutations, finding proliferation pathways enriched in basal and HER2 tumours and oestrogen response enriched in luminal tumours, cross-validated by the co-expression and differential expression results.

Computational Identification of Antimalarial Leads through Fingerprint Similarity Analysis

[GitHub Link](#)

- Screened a library of antimalarial molecules against a query (chloroquine drug) by generating molecular fingerprints and computing Tanimoto similarity coefficients using KNIME and RDKit nodes.
- Extracted molecular data from the Protein Data Bank (PDB) and visualized pairwise similarity scores as a heatmap to identify high-similarity candidates for antimalarial drug development.

In Silico Analysis of 3DTC/CEP-6331 Complex using Molecular Docking

[GitHub Link](#)

- Implemented a molecular docking simulation using AutoDock Vina to model the interaction between the 3DTC kinase and the CEP-6331 inhibitor, achieving a predicted binding affinity of -12.23 kcal/mol.
- Analyzed the top binding pose to identify the specific amino acids responsible for the stable interaction, including key hydrophobic (ILE180, LEU274) and polar (LYS228, ASP284) residues.

Single-Cell Transcriptomic Profiling of the B16 Melanoma's Microenvironment

[GitHub Link](#)

- Processed raw single-cell transcriptomic data from a B16 melanoma model using R (Seurat), applying quality control, normalization, and clustering to map the tumor's cellular heterogeneity.
- Characterized the resulting cell clusters by annotating fourteen distinct cell types within the tumor microenvironment and performed differential expression analysis to identify their unique gene signatures.

Honors and Leadership

First runner-up in the Biomimicry Innovation Challenge and Showcase, Vellore Institute of Technology, 2024

Top 10 Winner of Bio-Inspired Design Fest (BIDFEST) Ideathon, Vellore Institute of Technology, 2024

Vice Chairperson at Alpha Bio Cell, VIT University, India

January 2024 - December 2024

- Led several biotechnology and bioinformatics-related events, including hackathons, workshops, and quizzes, enhancing the club's engagement.
- Mentored junior core members and applied leadership, technical, and project management skills to ensure successful event planning and execution.

Extra-Curricular

Volunteer | National Service Scheme, NGO

Conducted sessions in government schools on career opportunities and the importance of joining the armed forces.

Volunteer | Becoming I Foundation, NGO

Taught science, math, and english in government schools, supporting underprivileged communities.

Event Coordinator | BioSummit, School of Biosciences and Technology

Organized a flagship event connecting academia and food industries, managing food distribution and ensuring smooth execution.